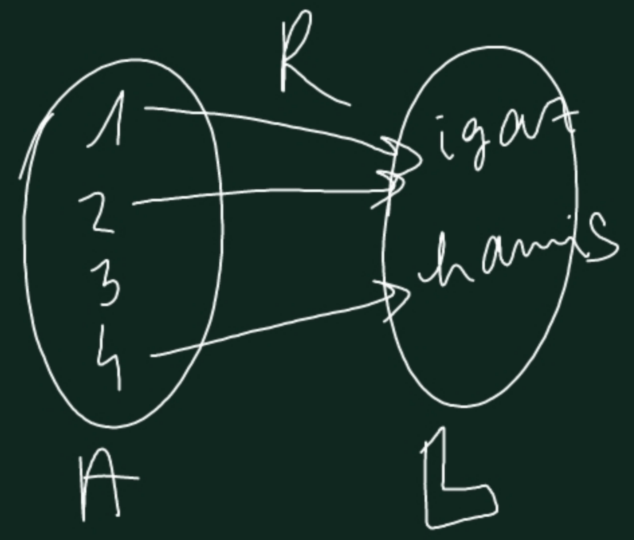
Logikai függvények

A legyen tetszőleges állapottér.

$$\boxed{R \in A \to \mathbb{L}}$$ R logikai függvény

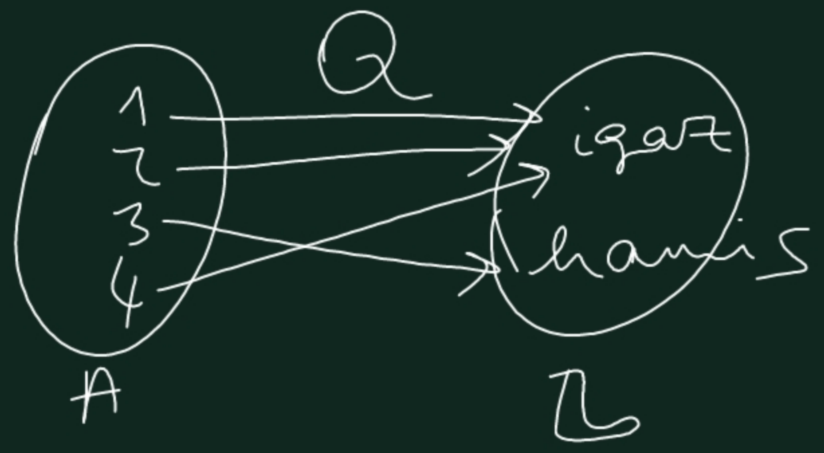
$$[R] = \{a \in A \,/\, R(a) = \{\text{igaz}\}\}$$

Példa



$$R = \{(1, \text{igaz}), (2, \text{igaz}), (4, \text{hamis})\}$$

$$[R] = \{1, 2\}$$



$Q : A \to \mathbb{Z}$

$Q = \{(1, \text{igaz}), (2, \text{igaz}), (3, \text{hamis}), (4, \text{igaz})\}$

$\lceil Q \rceil = \{1, 2, 4\}$

$IGAZ = \{(a, \text{igaz}) \mid a \in A\}$

$HAMIS = \{(a, \text{hamis}) \mid a \in A\}$

A ... tetsz. állapotHÉR

$\lceil IGAZ \rceil = A$

$\lceil HAMIS \rceil = \emptyset$

$\forall a \in A : IGAZ(a) = \text{igaz}$

$\forall a \in A : HAMIS(a) = \text{hamis}$

$A = [1..5]$

$Q, P : A \to \mathbb{Z}$

$\lceil P \rceil = \{1, 2\}$

$\lceil Q \rceil = \{1, 2, 3, 4\}$

$S(1) = \{\langle 1 \rangle, \langle 1, 2 \rangle\}$

$S(3) = \{\langle 3 \rangle, \langle 3, 4 \rangle, \langle 3, 3, 3, \ldots \rangle\}$

$D_{p(S)} = \{1, 2, 4, 5\}$ $(5, 5)$

$p(S)(1) = \{(1, 1), (1, 2)\}$

$p(S) = \{(1,1), (1,2), (2,2), (2,3), (4,4), (4,5)\}$

S program A felett

$$S = \{(a, \langle a \rangle) \mid a \in A\} \cup \{(a, \langle a, a+1 \rangle) \mid a \leq 4\} \cup$$

$$\{(a, \langle a, a, a, \ldots \rangle) \mid a = 3\}$$

Hány elemű S?

$2 + 2 + 3 + 2 + 1 = \underline{\underline{10}}$

$P \subseteq Q$?

$P = \{(1, \text{igor}), (2, \text{igor}), (3, \text{hanis}), (4, \text{hanis}), (5, \text{hanis})\}$

$Q = \{(1, \text{igor}), (2, \text{igor}), (3, \text{igor}), (4, \text{igor}), (5, \text{hanis})\}$

$(3, \text{hanis}) \notin Q$. <u>Hanis.</u>

Leggyengébb előfeltétel

Legyen A egy tetszőleges alapottér.

Legyen $R \in A \to \mathbb{L}$ logikai függvény.

Legyen $S \subseteq A \times (\bar{A} \cup \{\text{fail}\})^{**}$ program.

Az S program R utófeltételéhez tartozó legleggyengébb előfeltétel $lf(S,R) : A \to \mathbb{L}$ logikai függvény, amelyre teljesül, hogy $[lf(S,R)] = \{ a \in A \mid a \in Dp(S) \wedge p(S)(a) \subseteq [R] \}$.

$$\lceil lf(S,Q) \rceil = \{ a \in A \mid a \in \underbrace{Dp(S)}_{\{1,2,4,5\}} \wedge p(S)(a) \subseteq \underbrace{\lceil Q \rceil}_{\{1,2,3,4\}} \} =$$

$p(S)(1) = \{1,2\} \subseteq \lceil Q \rceil$

$p(S)(2) = \{2,3\} \subseteq \lceil Q \rceil$

$p(S)(4) = \{4,5\} \nsubseteq \lceil Q \rceil$

$p(S)(5) = \{5\} \nsubseteq \lceil Q \rceil$

$$= \{1,2\}$$

$4 \in \lceil lf(S,P) \rceil$?

$4 \in Dp(S)$? ✓ Здесь.

$\{4,5\} = p(S)(4) \nsubseteq \{1,2\} = \lceil P \rceil$ ✗ Нет.

$4 \notin \lceil lf(S,P) \rceil$.

A tetszőleges állapottér

$R : A \to \mathbb{B}$ logikai függvény

S program A felett

$$\lceil wp(SKIP, R) \rceil = \{ a \in A \mid a \in Dp(SKIP) \wedge p(SKIP)(a) \subseteq \lceil R \rceil \} =$$

A $\quad \{a\}$ $\quad a \in \lceil R \rceil$

$$= \{ a \in \lceil R \rceil \} = \lceil R \rceil$$

$$\lceil wp(ABORT, R) \rceil = \{ a \in A \mid a \in Dp(ABORT) \wedge p(ABORT)(a) \subseteq \lceil R \rceil \} =$$

$Dp(ABORT) = \emptyset$

$$= \emptyset$$

$$\lceil \mathcal{G}(S \mid HAMIS) \rceil = \{ a \in A \mid a \in D_{p(S)} \wedge p(S)(a) \subseteq \underbrace{\lceil HAMIS \rceil}_{= \emptyset} \} =$$

$$p(S)(a) \neq \emptyset$$
$$p(S)(a) = \emptyset$$

$$= \emptyset$$

$$\lceil \mathcal{G}(S \mid IGAZ) \rceil = \{ a \in A \mid a \in D_{p(S)} \wedge p(S)(a) \subseteq \underbrace{\lceil IGAZ \rceil}_{= A} \} =$$

$$= D_p(S)$$

A tetsz. állapotter

$R : A \to \mathbb{B}$ logikai függvény

S program A felett

Ellenpélda

$A = \{1, 2\}$

$[R] = \{2\}$ $\quad [\neg R] = \{1\}$

$S = \left\{ \begin{array}{l} 1 \to \langle 1, 1, 1, \ldots \rangle \\ 2 \to \langle 2 \rangle \end{array} \right\}$

ⓐ $[lf(S,R)] \cup [lf(S,\neg R)] = A$

$$[lf(S,R)] = \{a \in A \mid a \in D_{p(S)} \wedge p(S)(a) \subseteq [R]\}$$

$$= \{a \in A \mid a \in D_{p(S)} \wedge \forall b \in p(S)(a) : b \in [R]\}$$

$$[lf(S,\neg R)] = \{a \in A \mid a \in D_{p(S)} \wedge \forall b \in p(S)(a) : b \in [\neg R]\}$$

$$\cup \downarrow$$

$$\{a \in A \mid a \in D_{p(S)} \wedge (\forall b \in p(S)(a) : b \in [R]) \vee (\forall b \in p(S)(a) : b \in [\neg R])\} \subseteq$$

$$D_{p(S)} \subseteq A$$

ⓑ $[lf(S,R)] \cup [lf(S,\neg R)] = D_{p(S)}$

Hamis.

Hamis.

$$S = \left\{ \begin{array}{l} 1 \to \langle 1, 1, 1, \ldots \rangle \\ 2 \to \langle 2 \rangle \\ 2 \to \langle 2, 1 \rangle \end{array} \right\}$$

most

$$\lceil \mathcal{G}(S, R) \rceil = \emptyset$$

$$\lceil \mathcal{G}(S, \neg R) \rceil = \emptyset$$

$A = (x : \mathbb{N}, y : \mathbb{N})$

S

$\boxed{x := x - y}$

$(3,1) \longrightarrow \langle \underline{(3,1)}, (2,1) \rangle$

$(1,3) \longrightarrow \langle (1,3), \text{fail} \rangle$

$$p(S) = \{(a,b) \in A \times A \mid x(a) \geq y(a) \wedge y(b) = y(a) \wedge x(b) = x(a) - y(a)\}$$

$$D_{p(S)} = \{a \in A \mid S(a) \subseteq A^*\} = \{a \in A \mid x(a) \geq y(a)\}$$

$$\forall a \in D_{p(S)} : p(S)(a) = \{b \in A \mid y(b) = y(a) \wedge x(b) = x(a) - y(a)\}$$

$$R((x,y)) = (2x + y < 5)$$

$$R = \left\{ (a, \ell) \in A \times \mathbb{Z} \;/\; \ell = \begin{cases} \text{igaz, ha } 2x(a) + y(a) < 5 \\ \text{hamis, különben} \end{cases} \right\}$$

$$\boxed{\mathcal{U}(S,R)} = \left\{ a \in A \;/\; a \in D_{p(S)} \wedge p(S)(a) \subseteq \lceil R \rceil \right\} =$$

$$= \left\{ a \in A \;/\; x(a) \geq y(a) \wedge \forall b \in p(S)(a) : b \in \lceil R \rceil \right\} =$$

$$= \left\{ a \in A \;/\; x(a) \geq y(a) \wedge \forall b \in p(S)(a) : (x(b) = x(a) - y(a) \wedge y(b) = y(a) \right.$$

$$\left. \wedge\; 2 \cdot x(b) + y(b) < 5) \right\} =$$

$$= \left\{ a \in A \;/\; x(a) \geq y(a) \wedge 2 \cdot \boxed{(x(a) - y(a))} + \boxed{y(a)} < 5 \right\}$$

$\left(\begin{array}{l} x(b),\ y(b) - t \\ \text{helyettesítettük} \end{array} \right)$

$$\{ a \in A \mid x(a) \geq y(a) \wedge 2 \cdot (x(a) - y(a)) + y(a) < 5 \} =$$

$$= \{ a \in A \mid x(a) \geq y(a) \wedge 2x(a) - y(a) < 5 \}$$

$$\mathcal{U}(S, R) = (x \geq y \wedge 2x - y < 5)$$

$(2,2) \in \lceil \mathcal{U}(S,R) \rceil$ ✓

$(5,3) \in \lceil \mathcal{U}(S,R) \rceil$ X

$(2,2) \to \langle (2,2), (0,2) \rangle$

$(5,3) \to \langle (5,3), (2,3) \rangle$

A tetszőleges állapottér

$P, Q, Z : A \to \mathbb{L}$ logikai függvény

$$P \wedge \neg Q \Rightarrow \mathcal{G}(S, P \vee Q)$$

$$Q \Rightarrow Z$$

Ugyanis

$$P \wedge \neg Z \Rightarrow \mathcal{G}(S, P \vee Z)$$

$$P \wedge \neg Z \Rightarrow P \wedge \neg Q$$

$$P \wedge \neg Q \Rightarrow \mathcal{G}(S, P \vee Q)$$

$$\mathcal{G}(S, P \vee Q) \Rightarrow$$

$$\mathcal{G}(S, P \vee Z)$$

$Q \in A \to \mathbb{B}$

$R \in A \to \mathbb{B}$

$Q \Rightarrow R$, ha $\lceil Q \rceil \subseteq \lceil R \rceil$

$$\left(\left(\left(Q_{|\{a \in D_Q \mid Q(a) = \{iqa7\}\}} \subseteq R_{|\{a \in D_R \mid R(a) = \{iqa7\}\}} \subseteq R \right) \right) \right)$$